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Theme

AURA-MAS: Agentic and Multi-Agent Intelligent Surveillance

A Privacy-Aware, Edge-First Multi-Agent System for Multimodal Event Detection,

Inter-Sensor Coordination, and Explainable Alerting

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Abstract

Conventional video surveillance remains largely centralized and passive: streams from many cameras converge on a control room where overloaded human operators must notice rare critical events, a paradigm that scales poorly and reacts slowly. This thesis proposes and evaluates **AURA-MAS** (Agentic Understanding & Response Architecture), a privacy-aware, edge-first *hierarchical multi-agent system* for intelligent site surveillance that detects, corroborates, verifies, and explains security-relevant events with a human operator kept firmly in the loop.

At the edge layer, autonomous perception agents process video (YOLO11 detection with ByteTrack multi-object tracking, declarative zone rules, and CLIP-based zero-shot semantic anomaly scoring) and audio (sound event classification with a signal-processing fallback), publishing structured events—never raw footage—on a lightweight MQTT/Redis message bus. At the coordination layer, a fusion agent aggregates events into incident hypotheses using spatio-temporal late fusion with noisy-OR confidence combination; a coordinator agent resolves uncertain "gray-zone" hypotheses through auction-based task allocation inspired by the Contract Net Protocol, tasking the best-placed camera to actively verify the incident; and a deterministic policy agent applies auditable thresholds, severity mapping, and alert hysteresis. A rule-guarded *agentic* explanation layer built on large language models then transforms accepted alerts into grounded natural-language incident reports whose evidence citations are mechanically checked against the actual event log, ensuring that generative components can never fabricate evidence nor influence the alert decision itself.

The complete system is implemented in Python and evaluated on multi-sensor replay scenarios with ground-truth annotations. An architectural ablation shows that the distributed multi-agent design reduces mean time-to-alert by approximately 36% relative to a centralized sequential baseline, while auction-based coordination halves the false-alert rate compared to uncoordinated operation, at the cost of a modest, quantified message overhead. The thesis also contributes a reproducible evaluation methodology (event-level F1, time-to-alert, false alerts per hour, coordination overhead) and a privacy-by-design pipeline (edge-local processing, automatic anonymization of all exported evidence, and an immutable audit trail) aligned with GDPR principles and the risk-based logic of the EU AI Act.

Keywords— Intelligent Surveillance, Multi-Agent Systems, Agentic AI, Large Language Models, Object Detection, Multimodal Fusion, Auction-Based Coordination, Contract Net Protocol, Edge Computing, Explainable AI, Privacy by Design, Video Anomaly Detection

Résumé

La vidéosurveillance conventionnelle demeure largement centralisée et passive : les flux de nombreuses caméras convergent vers une salle de contrôle où des opérateurs surchargés doivent remarquer des événements critiques rares, un paradigme qui passe mal à l'échelle et réagit lentement. Cette thèse propose et évalue **AURA-MAS**, un *système multi-agents hiérarchique*, respectueux de la vie privée et orienté *edge*, pour la surveillance intelligente de sites : il détecte, corrobore, vérifie et explique les événements de sécurité tout en maintenant l'opérateur humain au centre de la décision.

Au niveau de la périphérie, des agents de perception autonomes traitent la vidéo (détection YOLO11 avec suivi multi-objets ByteTrack, règles de zones déclaratives et score d'anomalie sémantique zero-shot fondé sur CLIP) et l'audio (classification d'événements sonores avec repli en traitement du signal), en publiant des événements structurés — jamais de flux bruts — sur un bus de messages léger MQTT/Redis. Au niveau de la coordination, un agent de fusion agrège les événements en hypothèses d'incident par fusion tardive spatio-temporelle avec combinaison de confiance noisy-OR ; un agent coordinateur résout les hypothèses incertaines par allocation de tâches aux enchères inspirée du protocole Contract Net, en chargeant la caméra la mieux placée de vérifier activement l'incident ; et un agent de politique déterministe applique des seuils auditables, une cartographie de sévérité et une hystérésis d'alerte. Une couche d'explication *agentique* encadrée par des règles, fondée sur les grands modèles de langage, transforme ensuite les alertes acceptées en rapports d'incident en langage naturel dont les citations de preuves sont vérifiées mécaniquement par rapport au journal d'événements réel.

Le système complet est implémenté en Python et évalué sur des scénarios de rejeu multi-capteurs annotés. Une ablation architecturale montre que la conception multi-agents distribuée réduit le temps moyen d'alerte d'environ 36% par rapport à une base centralisée séquentielle, tandis que la coordination par enchères divise par deux le taux de fausses alertes par rapport à un fonctionnement non coordonné, au prix d'un surcoût de messages modeste et quantifié. La thèse contribue également une méthodologie d'évaluation reproductible et un pipeline de protection de la vie privée dès la conception (traitement local en périphérie, anonymisation automatique de toutes les preuves exportées et journal d'audit immuable), aligné sur les principes du RGPD et la logique fondée sur les risques de l'AI Act européen.

Mots-clés : Surveillance intelligente, Systèmes multi-agents, IA agentique, Grands modèles de langage, Détection d'objets, Fusion multimodale, Coordination par enchères, Protocole Contract Net, Informatique en périphérie, IA explicable, Protection de la vie privée dès la conception, Détection d'anomalies vidéo

ملخص

لا تزال المراقبة بالفيديو التقليدية مركزية وسلبية إلى حد كبير: تتدفق بثوث كاميرات عديدة نحو غرفة تحكم يتعين فيها على مشغلين مثقلين ملاحظة أحداث حرجة نادرة، وهو نموذج ضعيف التوسع وبطيء الاستجابة. تقترح هذه الأطروحة وتقيّم نظام AURA-MAS، وهو نظام متعدد الوكلاء هرمي يحترم الخصوصية ويعمل على حافة الشبكة، للمراقبة الذكية للمواقع: يكتشف الأحداث الأمنية ويعززها ويتحقق منها ويشرحها مع إبقاء المشغل البشري في قلب القرار.

على مستوى الحافة، يعالج وكلاء إدراك مستقلون الفيديو (كشف YOLO11 مع تتبع متعدد الأجسام ByteTrack وقواعد مناطق تصريحية ودرجة شذوذ دلالية بلا أمثلة قائمة على CLIP) والصوت (تصنيف الأحداث الصوتية مع بديل قائم على معالجة الإشارة)، وينشرون أحداثاً مهيكلة --- وليس تدفقات خام أبداً --- عبر ناقل رسائل خفيف MQTT/Redis. وعلى مستوى التنسيق، يجمّع وكيل الدمج الأحداث في فرضيات حوادث عبر دمج متأخر زماني-مكاني، ويحل وكيل التنسيق الفرضيات غير المؤكدة عبر تخصيص المهام بالمزاد المستوحى من بروتوكول Contract Net، بتكليف الكاميرا الأنسب بالتحقق النشط من الحادث، بينما يطبق وكيل السياسة الحتمي عتبات قابلة للتدقيق وخريطة خطورة وتباطؤ الإنذارات. ثم تحوّل طبقة شرح وكيلية مقيدة بالقواعد ومبنية على النماذج اللغوية الكبيرة الإنذارات المقبولة إلى تقارير حوادث بلغة طبيعية تُفحص استشهاداتها بالأدلة آلياً مقابل سجل الأحداث الفعلي.

نُفذ النظام الكامل بلغة بايثون وقُيّم على سيناريوهات إعادة تشغيل متعددة الحساسات ذات حقيقة أرضية. تُظهر دراسة الاستئصال المعمارية أن التصميم الموزع متعدد الوكلاء يقلل متوسط زمن الإنذار بنحو 36% مقارنة بخط أساس مركزي تسلسلي، بينما تخفض التنسيقات بالمزاد معدل الإنذارات الكاذبة إلى النصف مقارنة بالتشغيل غير المنسق، مقابل حمل رسائل متواضع ومقاس. كما تسهم الأطروحة بمنهجية تقييم قابلة لإعادة الإنتاج وخط أنابيب لحماية الخصوصية منذ التصميم (معالجة محلية على الحافة، إخفاء هوية تلقائي لجميع الأدلة المصدَّرة، وسجل تدقيق غير قابل للتغيير) بما يتماشى مع مبادئ اللائحة العامة لحماية البيانات ومنطق قانون الذكاء الاصطناعي الأوروبي القائم على المخاطر.

الكلمات المفتاحية: المراقبة الذكية، الأنظمة متعددة الوكلاء، الذكاء الاصطناعي الوكيلي، النماذج اللغوية الكبيرة، كشف الأجسام، الدمج متعدد الوسائط، التنسيق بالمزاد، بروتوكول Contract Net، الحوسبة الطرفية، الذكاء الاصطناعي القابل للتفسير، الخصوصية منذ التصميم، كشف الشذوذ في الفيديو.

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Acronyms

ACL Agent Communication Language. 17

BDI Belief–Desire–Intention. 17

CLIP Contrastive Language–Image Pre-training. 21, 29, 45, 48

CNP Contract Net Protocol. 13, 17, 30

DSP Digital Signal Processing. 21, 30, 35, 41, 48

FIPA Foundation for Intelligent Physical Agents. 17

FPS Frames Per Second. 29, 40

GDPR General Data Protection Regulation. 14, 24

IoT Internet of Things. 17, 23

IoU Intersection over Union. 30

JSON JavaScript Object Notation. 17, 29, 31, 32, 35, 36

LLM Large Language Model. 12, 13, 18, 24, 27, 35, 48

mAP mean Average Precision. 20, 39

MARL Multi-Agent Reinforcement Learning. 18, 23, 49

MAS Multi-Agent System. 12–14, 17–19, 27, 42, 45

MQTT Message Queuing Telemetry Transport. 13, 17, 23, 28, 48

QoS Quality of Service. 23, 28, 32

Re-ID Person Re-Identification. 14, 32

SED Sound Event Detection. 21, 24, 48

VAD Video Anomaly Detection. 12, 21, 22, 24

VLM Vision-Language Model. 12, 13, 18, 24, 27, 31

YOLO You Only Look Once. 12, 20

Part I

Introduction

Chapter 1

General Introduction

1.1 Context and Motivation

Video surveillance has become one of the most ubiquitous applications of computer vision, with hundreds of millions of cameras deployed across cities, campuses, industrial facilities, and transport infrastructure. Yet the dominant operating paradigm has changed remarkably little in three decades: camera streams converge on a centralized control room, where a small number of human operators are expected to monitor dozens of screens simultaneously and to notice rare, safety-critical events in real time. Empirical studies of operator performance consistently show that sustained attention over multiple video feeds degrades within minutes, that the ratio of cameras to operators makes exhaustive monitoring physically impossible, and that most recorded footage is only ever examined *after* an incident, for forensic purposes rather than prevention.

The first generation of "intelligent" video analytics attempted to relieve this burden with centralized detection pipelines: motion detection, later deep-learning-based object detection and video anomaly detection (Video Anomaly Detection (VAD)), running on servers that ingest all streams. While detection accuracy has improved dramatically—modern detectors such as You Only Look Once (YOLO) achieve real-time performance with high mean average precision, and weakly-supervised VAD methods reach strong results on benchmarks such as UCF-Crime [1]—the centralized architecture itself has become the bottleneck. It concentrates bandwidth and compute demands, creates a single point of failure, raises serious privacy concerns because raw footage of people is transported and stored centrally, and produces floods of uncorrelated, unexplained alerts that overwhelm rather than assist the operator.

In parallel, two research currents have matured to the point where a fundamentally different architecture becomes practical. First, *multi-agent systems* (Multi-Agent System (MAS)) provide a principled framework for distributing perception and decision-making across autonomous, cooperating software agents, with well-studied coordination mechanisms such as the Contract Net Protocol [2] and auction-based task allocation [3]. Second, the recent emergence of *agentic AI*—systems in which large language models (Large Language Model (LLM)s) and vision-language models (Vision-Language Model (VLM)s) plan, use tools, and reason over multimodal evidence [4, 5]—makes it possible to add a semantic layer that can interpret, corroborate, and *explain* what the perception layer detects, in natural language a human operator can act upon.

This thesis sits precisely at the intersection of these currents. It asks: *how should perception, coordination, decision, and explanation be organized in a modern surveillance system so that the system is faster, more accurate, more private, and more trustworthy than a centralized pipeline?*

1.2 Problem Statement

The problem addressed in this thesis can be stated as follows:

Design, implement, and evaluate a multi-agent intelligent surveillance system for a semi-closed site, in which autonomous perception agents detect security-relevant events from video and audio at the edge; coordinate with one another to corroborate and actively verify uncertain detections; apply deterministic, auditable alerting policies; and generate grounded natural-language explanations for a human operator—all under privacy-by-design constraints.

This problem decomposes into five research questions:

- **RQ1 (Architecture).** Does a distributed, hierarchical MAS architecture measurably improve responsiveness (time-to-alert) and scalability over a centralized sequential pipeline processing the same sensor streams?
- **RQ2 (Coordination).** Does explicit inter-agent coordination—specifically auction-based verification task allocation inspired by the Contract Net Protocol (CNP)—improve alert precision compared to uncoordinated per-sensor alerting, and at what communication cost?
- **RQ3 (Multimodality).** Does fusing audio events with video events increase detection confidence and reduce false alerts compared to vision-only operation?
- **RQ4 (Agentic explanation).** Can an LLM-based agentic layer generate incident explanations that are strictly grounded in system evidence—i.e., provably free of fabricated references—without influencing or delaying the safety-critical alert decision?
- **RQ5 (Governance).** Can the entire pipeline operate under privacy-by-design constraints (no raw footage leaves the edge, all exported evidence anonymized, all decisions audited) while remaining useful to the operator?

1.3 Objectives and Contributions

The overall objective is to deliver a working, evaluated prototype—named **AURA-MAS** (Agentic Understanding & Response Architecture — Multi-Agent System)—together with a reproducible evaluation methodology. The contributions of this thesis are:

C1 — A hierarchical multi-agent surveillance architecture. A three-layer design (edge perception, site coordination, governance/operator) with seven specialized agent roles, a formal event/alert message schema, and a lightweight Message Queuing Telemetry Transport (MQTT)/Redis communication substrate (Chapter 5).

C2 — An auction-based active verification protocol. A single-round auction mechanism, derived from the CNP, in which uncertain "gray-zone" incident hypotheses trigger a verification task auctioned among camera agents; the winning agent performs a high-scrutiny re-analysis whose outcome adjusts the fused confidence before the alert decision (Chapters 5–6).

C3 — Multimodal spatio-temporal late fusion. A noisy-OR confidence combination scheme with per-modality reliability weights and cross-modal corroboration bonuses, which guarantees that independent supporting evidence strictly increases incident confidence (Chapter 5).

C4 — A rule-guarded agentic explanation layer. An LLM/VLM agent that produces natural-language incident reports *after* the deterministic alert decision, with a mechanical guardrail that verifies every cited evidence identifier against the actual event log and falls back to a deterministic template on any violation (Chapter 6).

C5 — A reproducible evaluation methodology and ablation study. Scenario replay infrastructure with ground-truth manifests, system-level metrics (event-level F1, mean time-to-alert, false alerts per hour, coordination overhead), and a four-way architectural ablation: centralized baseline, MAS without coordination, MAS with rule-based scheduling, and MAS with auction coordination (Chapter 7).

1.4 Scope and Assumptions

The thesis targets *semi-closed sites*—campuses, warehouses, industrial facilities—where the set of cameras and microphones is known, zones can be declared (restricted areas, entries), and a single security operator team is responsible. The system performs *event detection and alerting*, not person identification: no facial recognition, biometric matching, or person re-identification (Person Re-Identification (Re-ID)) across time is performed, by deliberate design choice aligned with the General Data Protection Regulation (GDPR) data-minimization principle [6] and the prohibitions and high-risk provisions of the EU AI Act [7]. Evaluation is performed by replaying recorded multi-sensor scenarios with ground-truth annotations rather than live deployment, which enables controlled, repeatable ablations.

1.5 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 introduces the theoretical foundations of multi-agent systems and agentic AI. Chapter 3 covers the perception building blocks: object detection, multi-object tracking, video anomaly detection, and audio event detection. Chapter 4 surveys the state of the art in distributed and multi-agent surveillance, agentic orchestration frameworks, and the legal/ethical governance landscape, and positions our contribution. Chapter 5 presents the design of AURA-MAS: architecture, agent roles, message schemas, fusion and coordination mechanisms, and privacy-by-design measures. Chapter 6 details the implementation: technology stack, code organization, and the algorithms of each agent. Chapter 7 describes the experimental setup, metrics, results of the architectural ablation, and discusses findings and limitations. Chapter 8 concludes and outlines future work.

Part II

Background and State of the Art

Chapter 2

Multi-Agent Systems and Agentic AI: Foundations

2.1 Introduction

This chapter establishes the conceptual vocabulary used throughout the thesis. We first define agents and multi-agent systems, then review the coordination mechanisms most relevant to distributed surveillance—in particular task allocation by auction—and finally introduce the recent paradigm of *agentic AI*, in which large language models act as reasoning engines inside agent architectures.

2.2 Agents and Multi-Agent Systems

2.2.1 The Agent Abstraction

Following Wooldridge [8], an **agent** is a computer system situated in an environment, capable of *autonomous action* in that environment in order to meet its design objectives. Four properties are commonly required:

- **Autonomy**: the agent operates without direct external control of its internal state and decisions;
- **Reactivity**: it perceives its environment and responds in a timely fashion to changes;
- **Pro-activeness**: it exhibits goal-directed behavior, taking initiative rather than only reacting;
- **Social ability**: it interacts with other agents through some communication language or protocol.

In AURA-MAS, a camera perception agent exemplifies all four: it autonomously processes its own video stream (autonomy), emits events when zone rules fire (reactivity), continuously maintains tracking state and dwell-time counters (pro-activeness), and bids on verification tasks announced by a coordinator (social ability).

2.2.2 Agent Architectures

Classical agent architectures span a spectrum from purely *reactive* (stimulus–response rules, subsumption) to *deliberative* (symbolic planning over explicit world models), with the influential

Belief–Desire–Intention (Belief–Desire–Intention (BDI)) model [9] occupying a pragmatic middle ground: agents maintain *beliefs* about the world, adopt *desires* (goals), and commit to *intentions* (plans). Hybrid layered architectures combine a fast reactive layer with a slower deliberative layer—a pattern that AURA-MAS mirrors at the system level: edge agents are predominantly reactive (frame-rate perception loops), while the coordination layer deliberates over aggregated hypotheses at a slower cadence, and the agentic explanation layer performs the slowest, most expensive reasoning only when justified.

2.2.3 Multi-Agent Systems

A **multi-agent system** (MAS) is a collection of interacting agents that jointly solve problems beyond the capability or knowledge of any individual agent [8]. MAS are attractive for surveillance for several structural reasons:

- **Natural distribution**: sensors are physically distributed; assigning one agent per sensor matches the problem topology and keeps raw data local.
- **Scalability**: adding a camera means adding an agent, not re-dimensioning a central server.
- **Robustness**: the failure of one agent degrades coverage locally instead of collapsing the whole system.
- **Heterogeneity**: video, audio, and higher-level reasoning agents can coexist behind a common message interface.

Interaction is standardized by agent communication languages; the Foundation for Intelligent Physical Agents (FIPA) Agent Communication Language (ACL) defines performatives (`inform`, `request`, `cfp`, `propose`, `accept-proposal`, …) whose semantics underpin most practical coordination protocols. AURA-MAS adopts the *spirit* of FIPA interaction protocols while using lightweight JavaScript Object Notation (JSON) messages over MQTT topics, a common engineering compromise in modern Internet of Things (IoT)-oriented deployments [10].

2.3 Coordination and Task Allocation

2.3.1 The Coordination Problem

Coordination is the process of managing dependencies between agents' activities. In distributed surveillance the canonical dependency is *verification*: when one sensor produces an uncertain detection, which other sensor should spend resources to confirm or refute it? This is an instance of the *multi-robot/multi-agent task allocation* problem, for which three broad families of solutions exist.

2.3.2 The Contract Net Protocol and Auctions

The Contract Net Protocol (CNP) introduced by Smith [2] structures task allocation as a negotiation: a *manager* announces a task (call for proposals), *contractors* evaluate it against their local capabilities and submit *bids*, and the manager awards the task to the best bidder. Market-based extensions formalize bids as utilities and prove strong properties for single-item auctions [3]: allocation quality equals the best available bid, communication cost is linear in the number of bidders, and the mechanism degrades gracefully when agents fail to respond.

Auctions are particularly well suited to camera verification tasks because a camera's utility for verifying an event is inherently *local, private knowledge*—its field-of-view overlap with the event zone, its current processing load, whether it originated the initial detection—which the bidder can compute cheaply and truthfully summarize as a scalar bid. AURA-MAS therefore adopts a single-round, sealed-bid auction (Section 5.7) with a fixed bidding window, and retains a round-robin scheduler as an ablation baseline.

2.3.3 Alternative Approaches

Two alternatives deserve mention. *Centralized optimization* (e.g., Hungarian assignment over a global utility matrix) yields optimal allocations but requires the center to know all utilities, reintroducing the communication and single-point-of-failure costs that MAS avoid; recent work explores submodular optimization for multi-camera collaborative perception [11]. *Multi-agent reinforcement learning* (Multi-Agent Reinforcement Learning (MARL)) learns coordination policies end-to-end, with algorithms such as QMIX [12] and MADDPG [13] demonstrating strong results in simulated cooperative sensing and patrolling [14, 15]; standardized environments (PettingZoo [16]) and scalable training stacks (RLlib [17]) exist. However, MARL policies are data-hungry, hard to certify, and opaque—undesirable properties for a safety- and legally-sensitive system. We therefore position MARL-based active sensing as future work (Chapter 8) and keep the deployed coordination mechanism auditable.

2.4 Agentic AI

2.4.1 From Language Models to Agents

Large language models exhibit emergent capabilities for planning, tool use, and multi-step reasoning. The *agentic AI* paradigm operationalizes these capabilities: an LLM is embedded in a control loop where it observes state, reasons (often via explicit "thought" traces as in ReAct [5]), invokes tools, and iterates. Sapkota et al. [4] distinguish *AI agents*—single-model, tool-using systems—from *agentic AI*: orchestrated ensembles of specialized agents with shared memory and explicit coordination, which is precisely the regime of interest for surveillance, where perception specialists must feed reasoning specialists.

2.4.2 Orchestration Frameworks

A rich ecosystem of frameworks now supports multi-agent LLM orchestration: AutoGen structures multi-agent conversations [18]; LangGraph models agent workflows as typed state graphs with explicit nodes, edges, and checkpointing [19]; CrewAI emphasizes role-based teams. For vision-centric pipelines, research systems such as VideoAgent [20] and Hawk [21] demonstrate LLM-orchestrated video understanding, while VLMs such as GPT-4V [22], Gemini [23], Qwen-VL [24], and LLaVA [25] provide the underlying multimodal perception-to-language capability.

2.4.3 Reliability and Guardrails

The central obstacle to deploying generative models in safety-critical loops is *hallucination*: fluent but unfounded output. Guardrail frameworks such as NeMo Guardrails [26] interpose programmable checks between the model and the application. The design stance adopted in this thesis is stricter and architectural: the agentic layer is *decision-decoupled*—it runs only after a deterministic policy engine has already made the alert decision—and *evidence-grounded*—every evidence identifier cited in a generated report is mechanically checked against the event log, with automatic fallback to a template on violation. This turns the LLM from a potential safety liability into a strictly additive usability layer, a pattern we consider one of the transferable lessons of this work.

2.5 Conclusion

This chapter defined the agent abstraction, motivated the MAS architecture for distributed surveillance, selected auction-based task allocation as the coordination mechanism on grounds of efficiency and auditability, and framed agentic AI as a rule-guarded explanation layer rather than a decision maker. The next chapter reviews the perception techniques that edge agents rely upon.

Chapter 3

Perception for Intelligent Surveillance

3.1 Introduction

Every surveillance decision ultimately rests on perception: converting raw pixels and audio samples into structured, semantically meaningful events. This chapter reviews the four perception capabilities used by AURA-MAS edge agents—object detection, multi-object tracking, video anomaly detection, and audio event detection—with emphasis on the accuracy/latency trade-offs that govern edge deployment.

3.2 Object Detection

Object detection localizes and classifies objects in images. Modern real-time detection is dominated by the YOLO family, which frames detection as a single dense regression pass over the image. The current generation (YOLO11, released by Ultralytics in 2024 [27]) provides a spectrum of model scales; the nano variant (YOLO11n, ≈2.6M parameters) achieves approximately 39.5% mean Average Precision (mAP) on COCO while sustaining real-time throughput on CPU and embedded hardware, making it the natural choice for per-camera edge agents. Transformer-based detectors such as RT-DETR [28] eliminate non-maximum suppression and reach higher accuracy at moderate additional cost; they constitute a drop-in upgrade path in our architecture since agents encapsulate their detector behind a common event interface.

Two practical considerations matter for surveillance specifically. First, detection confidence thresholds trade recall against precision at the *event* level, not just the box level: a missed person is a missed intrusion. Second, pretrained COCO classes (person, car, truck, backpack, …) cover the vast majority of surveillance-relevant objects, so fine-tuning is optional rather than blocking—an important de-risking property for a time-boxed project [29].

3.3 Multi-Object Tracking

Detection alone yields per-frame boxes; surveillance rules (loitering, line crossing, abandonment) require *identity over time*. Multi-object tracking (MOT) associates detections across frames into trajectories. The prevailing paradigm is tracking-by-detection, and the reference algorithm is ByteTrack [30], whose key insight is to associate *every* detection box—including low-confidence ones—in a two-stage matching (high-confidence first, then low-confidence against unmatched tracks), which markedly reduces identity switches under occlusion. ByteTrack is integrated natively in the Ultralytics runtime (`model.track(..., tracker="bytetrack.yaml")`), which AURA-MAS uses directly. Evaluation of trackers uses HOTA/MOTA/IDF1 metrics computed by tools such as TrackEval [31, 32]; multi-camera extensions of tracking (cross-camera re-identification) exist [33, 34] but are deliberately excluded from this thesis on privacy grounds—our cross-camera corroboration operates at the *event* level, not the identity level.

3.4 Video Anomaly Detection

VAD aims to flag frames or segments that deviate from normal patterns. Three families coexist [35]:

- **Reconstruction/prediction methods** (autoencoders, future-frame prediction) trained on normal data only; effective on constrained scenes (Avenue, ShanghaiTech) but brittle under domain shift.
- **Weakly-supervised methods**, trained with video-level labels on datasets such as UCF-Crime [1]; the current accuracy leaders.
- **Zero-shot semantic methods**, which exploit vision-language alignment: a frame embedding is compared against natural-language descriptions of normal versus anomalous situations using Contrastive Language–Image Pre-training (CLIP) [36]. Recent variants (VadCLIP-style, AVadCLIP with audio [37], training-free EventVAD [38], explainable EX-VAD [39]) show that useful anomaly signals can be obtained *without any training*, which is exceptionally attractive for a one-month project.

AURA-MAS uses the third family as an optional semantic scorer on sampled frames: anomaly score is the softmax mass assigned to a set of anomalous text prompts ("people fighting", "fire and smoke", …) relative to normal prompts. This complements—rather than replaces—the deterministic zone-rule engine, and every CLIP-triggered event is subject to the same downstream fusion, verification, and policy thresholds as any other event.

3.5 Audio Event Detection

Audio is a cheap, omnidirectional, lighting-invariant modality that video-only systems ignore. Sound event detection (Sound Event Detection (SED)) classifies acoustic events—glass breaking, screams, gunshots, alarms—from short audio windows. Pretrained models dominate practice: YAMNet (MobileNet-based, 521 AudioSet classes) [40, 41], PANNs [42], the Audio Spectrogram Transformer [43], and BEATs [44] offer increasing accuracy at increasing cost; efficient distillations exist for edge deployment [45, 46]. A surveillance-relevant subset of classes can be mapped directly to alert-worthy event types, with per-class confidence thresholds. Because deep audio stacks can be heavy for constrained hardware, AURA-MAS additionally implements a dependency-light Digital Signal Processing (DSP) fallback: rolling z-scores of short-time energy and spectral flatness flag acoustic transients as generic `audio_anomaly` events—sufficient to demonstrate multimodal fusion even where TensorFlow is unavailable.

3.6 Multimodal Fusion

Fusion strategies are classified as early (feature-level), late (decision-level), or hybrid. For a distributed system, *late fusion* is the natural fit: each agent emits calibrated event confidences, and a fusion node combines them without needing access to raw features—preserving both bandwidth and privacy. AURA-MAS groups events into spatio-temporal hypotheses (same incident family, same zone, within a sliding window) and combines confidences with a noisy-OR model weighted by per-modality reliability, plus small calibrated bonuses for cross-modality and cross-sensor corroboration

(Section 5.6). Audio-visual VAD research confirms that such corroboration materially improves robustness [37, 47].

3.7 Datasets and Evaluation Data

Public datasets relevant to this work include UCF-Crime (1,900 untrimmed real-world videos across 13 anomaly classes) [1], Avenue and ShanghaiTech for scene-specific VAD, VIRAT for activity detection from static surveillance cameras, and MEVA for large-scale multi-camera activity detection [48]. Synthetic data generation is an increasingly credible complement [49, 50, 51]. For *system-level* evaluation (as opposed to model-level), no standard benchmark exists that combines multi-sensor streams with ground-truth incident annotations; this thesis therefore contributes replayable scenario manifests built from public clips and scripted recordings (Chapter 7), following the methodological lead of ETISEO [52] and modern MLOps evaluation practice [53].

3.8 Conclusion

The perception toolbox for surveillance is mature: pretrained real-time detectors, robust trackers, zero-shot semantic anomaly scorers, and pretrained audio classifiers can be composed without any model training. The engineering challenge—and the scientific opportunity—has shifted upward, to the architecture that organizes these components: how their outputs are fused, verified, decided upon, and explained. That architecture is the subject of the remainder of this thesis.

Chapter 4

State of the Art in Distributed and Agentic Surveillance

4.1 Introduction

This chapter surveys the four research areas that AURA-MAS synthesizes—distributed/edge video analytics, multi-agent and MARL-based surveillance, agentic multimodal alerting, and privacy/legal governance—and concludes by positioning our contribution against the identified gaps.

4.2 Distributed and Edge Video Analytics

The migration of video analytics from cloud to edge is driven by bandwidth, latency, and privacy [54, 55]. Reference industrial stacks (NVIDIA DeepStream [56]) demonstrate multi-stream, hardware-accelerated inference at the edge, while cross-platform runtimes (ONNX Runtime [57]) and single-board computers make per-camera inference economically viable [58]. Edge-cloud collaborative architectures partition work by confidence: cheap models filter at the edge, escalating ambiguous cases upward [59]—a pattern AURA-MAS generalizes into agent-level escalation (edge perception → site coordination → agentic explanation). Studies of distributed surveillance systems confirm architectural benefits in elasticity and fault containment [60, 61].

On the communication substrate, comparative studies of IoT messaging protocols consistently find MQTT superior for constrained, event-driven workloads [62, 63, 64], with Mosquitto as the standard broker [10] and Quality of Service (QoS) levels providing per-topic delivery guarantees. Real-time alert pipelines increasingly pair a pub/sub bus with a durable stream store (Redis Streams / Kafka) for alerts and audit [65, 66, 67]. AURA-MAS adopts exactly this two-tier substrate.

4.3 Multi-Agent and Learning-Based Surveillance

Classical multi-camera coordination research addresses camera hand-off, field-of-view negotiation, and distributed tracking [33, 68]. Task allocation via market mechanisms remains the practical workhorse [2, 3], with modern variants addressing task-oriented communication under bandwidth constraints [69] and submodular sensor selection [11].

The learning-based line applies MARL to cooperative sensing: multi-UAV cooperative search [14], cooperative target coverage [15], and improved coverage formulations [70] demonstrate that learned policies can outperform hand-crafted coordination in simulation. Value-decomposition methods (QMIX [12]) and centralized-critic methods (MADDPG [13]) are the standard algorithms, supported by PettingZoo environments [16] and RLlib [17]. However, published deployments in real surveillance are scarce: sample complexity, sim-to-real gaps, and non-auditable policies remain blocking for safety-critical use—a gap our auditable auction design deliberately sidesteps.

4.4 Agentic and Multimodal Alerting

The newest current applies LLM/VLM agents to video understanding. Hawk [21] and VideoAgent [20] orchestrate perception tools from an LLM planner; AVadCLIP [37] and related audio-visual methods [47] fuse modalities for anomaly detection; description-based SED [71] and explainable VAD [39, 72] target human-interpretable outputs; multi-agent LLM frameworks (AutoGen [18], LangGraph [19], surveyed in [4, 73, 74]) provide orchestration machinery, with resource-efficient serving studied in [75]. Reliability remains the open problem: guardrail systems [26] and LLM-as-judge evaluation [76] mitigate but do not eliminate hallucination. Notably, none of the surveyed systems architecturally *decouples* the generative component from the alert decision; they either use the LLM as the detector itself or as an unchecked narrator.

4.5 Privacy, Ethics, and Legal Governance

Surveillance AI is now heavily regulated territory. The GDPR imposes lawfulness, data minimization, and storage limitation on video processing [6]; European guidance discourages centralized retention of identifiable footage [77, 78]. The EU AI Act [7] prohibits real-time remote biometric identification in public spaces (with narrow exceptions) and classifies most other surveillance AI as high-risk, mandating risk management, logging, human oversight, and transparency [79, 80, 81, 82]. Trustworthy-AI frameworks (NIST AI RMF [83], reviewed in [84, 85]) converge on the same operational requirements: edge-local processing, anonymization, audit trails, and human-in-the-loop decision authority. These requirements are *constitutive* design constraints for AURA-MAS, not afterthoughts.

4.6 Synthesis and Positioning

Table 4.1 summarizes the gap analysis.

Table 4.1: Gap analysis: existing system families versus AURA-MAS.

Capability	**Centralized VMS analytics**	**MARL surveillance (research)**	**LLM video agents**	**AURA-MAS (this thesis)**
Edge-local perception	Partial	Simulated	No (cloud VLM)	Yes
Explicit coordination	No	Learned, opaque	Conversational	Auction (auditable)
Multimodal fusion	Rare	Rare	Partial	Yes (noisy-OR late fusion)
Explainable alerts	No	No	Yes, ungrounded	Yes, evidence-checked
Decision/generation decoupling	—	—	No	Yes (policy-first)
Privacy by design	Weak	Not addressed	Not addressed	Constitutive
System-level evaluation	Vendor claims	Simulation reward	Qualitative	Replayable ablations

Three gaps emerge. **(G1)** No surveyed system combines auditable multi-agent coordination with modern zero-shot perception. **(G2)** Agentic explanation layers are never decision-decoupled and evidence-grounded by construction. **(G3)** System-level evaluation methodology (time-to-alert, false alerts per hour, coordination overhead, under controlled ablation) is largely absent from the academic literature, which evaluates models rather than systems. AURA-MAS addresses all three, at prototype scale, within explicit legal design constraints.

4.7 Conclusion

The state of the art provides mature components and clear regulatory constraints but no integrated, auditable, privacy-preserving multi-agent architecture with grounded agentic explanation. The next chapter presents our design.

Part III

Contribution: Design of AURA-MAS

Chapter 5

AURA-MAS: System Design

5.1 Introduction

This chapter presents the design of AURA-MAS, from global architecture down to message schemas and coordination protocols. The design is governed by five principles derived from the analysis of Chapters 2–4:

1. **Edge-first**: raw sensor data is processed where it is produced and never leaves the edge agent;
2. **Events, not streams**: all inter-agent communication consists of small structured messages;
3. **Deterministic decisions**: alerts are produced exclusively by an auditable rule-based policy engine;
4. **Guarded generativity**: LLM/VLM components explain decisions but can never make or alter them;
5. **Privacy by design**: anonymization, minimization, and auditability are structural, not optional.

5.2 Global Architecture

AURA-MAS is a hierarchical MAS organized in three layers connected by a two-tier message substrate (Figure 5.1).

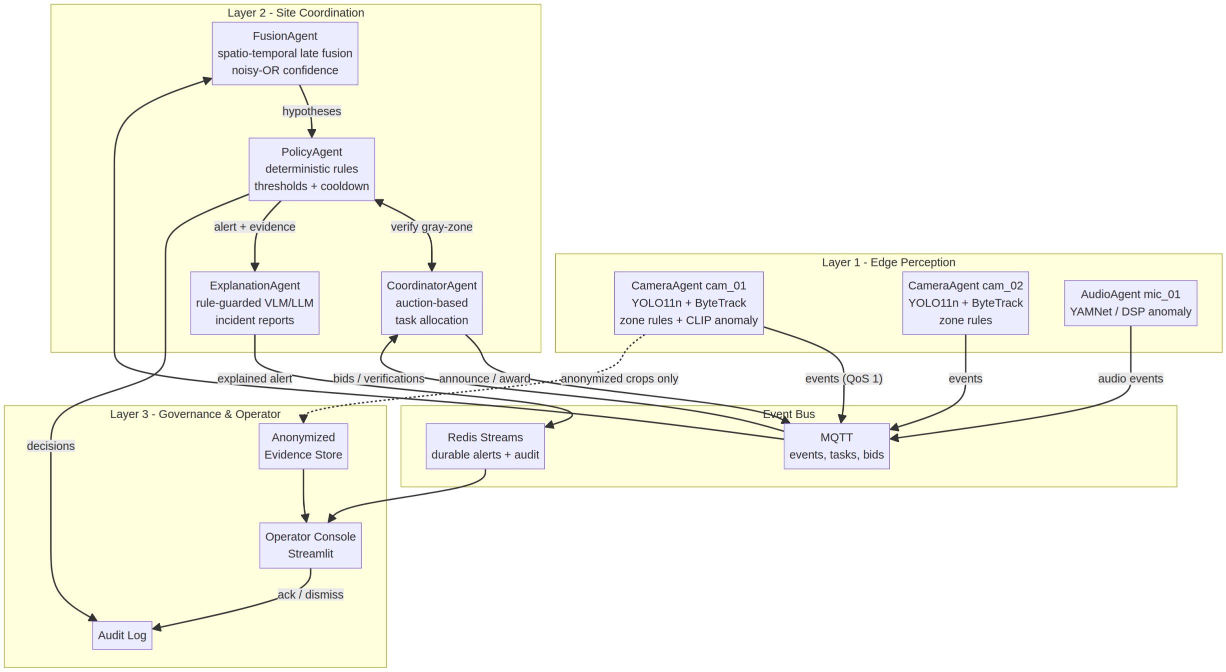

Figure 5.1: AURA-MAS architecture: edge perception layer, site coordination layer, and governance/operator layer, connected by an MQTT event bus and a Redis durable stream.

Layer 1 — Edge perception. One autonomous agent per sensor. *CameraAgents* run detection, tracking, zone rules, and optional semantic anomaly scoring on their own video stream. *AudioAgents* classify sound events or score acoustic anomalies. Both publish structured events on the bus.

Layer 2 — Site coordination. The *FusionAgent* aggregates events into incident hypotheses; the *CoordinatorAgent* allocates verification tasks by auction when hypotheses are uncertain; the *PolicyAgent* converts hypotheses into alerts (or suppressions) under deterministic rules; the *ExplanationAgent* generates grounded natural-language reports for accepted alerts.

Layer 3 — Governance and operator. A web operator console presents the alert feed, anonymized evidence, and explanations; operator actions (acknowledge/dismiss) and all policy decisions are appended to an immutable audit log.

The substrate pairs MQTT [10] for low-latency fan-out of high-frequency messages (detections QoS 0; events, tasks, bids, awards QoS 1) with Redis Streams for durable, replayable storage of alerts and audit records. A single-process in-memory bus with identical semantics supports development and testing.

5.3 Agent Roles and Responsibilities

Table 5.1: Agent roles in AURA-MAS.

Agent	Responsibility	Consumes	Produces
CameraAgent	Detection, tracking, zone rules, CLIP anomaly, verification bids/execution	video frames; tasks, awards	detections, events, bids, verifications
AudioAgent	Sound event classification / DSP anomaly	audio chunks	events
FusionAgent	Spatio-temporal hypothesis formation, confidence fusion	events	hypotheses
CoordinatorAgent	Auction-based verification task allocation	hypotheses (gray-zone), bids, verifications	tasks, awards
PolicyAgent	Severity mapping, thresholds, hysteresis, alert emission, audit	hypotheses, verification results	alerts, audit records
ExplanationAgent	Grounded incident report generation with guardrails	alerts, hypotheses, evidence	explanations
Operator console	Human oversight, acknowledgment	alerts, evidence	operator audit records

5.4 Message Schemas

All messages are JSON documents with mandatory provenance fields. The three core schemas are:

Detection (high-frequency, transient): `{sensor_id, frame_id, timestamp, objects: [{class, confidence, bbox, track_id}]}`.

Event (semantic, durable): `{event_id, sensor_id, timestamp, event_type, confidence` $\in [0,1]$`, modality, zone, track_id, evidence_path, extra}`. Event types include `intrusion`, `loitering`, `abandoned_object`, `anomaly`, and audio types (`audio_glass_break`, `audio_scream`, …). The `evidence_path` always references an *anonymized* crop.

Alert (operator-facing): `{alert_id, timestamp, severity` $\in$ `{CRITICAL, WARNING, INFO}, event_type, confidence, zone, sensors[], evidence[], fused_events[], explanation, status}`.

5.5 Edge Perception Design

5.5.1 Camera Pipeline

Each CameraAgent samples its stream at a configurable inference rate (default 5 Frames Per Second (FPS)) and runs a four-stage pipeline: (1) YOLO11n detection with ByteTrack association [27, 30]; (2) a stateful *zone rule engine* evaluating declarative site rules; (3) optional CLIP zero-shot anomaly scoring on subsampled frames [36]; (4) event emission with anonymized evidence.

The zone rule engine implements three rules over tracked objects, using the foot point (bottom-center of the bounding box) for zone membership tested by ray-casting against declared polygons:

- **Intrusion**: a `person` track enters a zone declared `restricted`; fired once per (track, zone) pair.
- **Loitering**: dwell time of a person track inside any zone exceeds a threshold τ_{loiter} (default 8 s); confidence grows with dwell time.
- **Abandoned object**: a non-person track remains static (Intersection over Union (IoU) with its initial box > 0.6) longer than τ_{aband} (default 10 s).

5.5.2 Audio Pipeline

AudioAgents process audio in overlapping windows extending up to 1.5 s of lookback context per 1-second chunk, so that short transient sounds are scored with sufficient temporal context rather than against a bare 1 s window. In YAMNet mode[1] [40], a curated mapping from AudioSet classes to surveillance event types (with per-class thresholds) produces typed events. In DSP fallback mode, rolling z-scores of short-time energy and spectral flatness produce generic `audio_anomaly` events, normalized to $[0, 1]$.

5.6 Fusion Design

The FusionAgent maintains a set of open *hypotheses*, keyed by (incident family, zone), where families group mutually corroborating event types (e.g., `intrusion` and `audio_glass_break` both belong to *security*). An incoming event either joins the open hypothesis for its key (if within the sliding window W, default 6 s, of the last event) or opens a new one.

Hypothesis confidence uses a reliability-weighted noisy-OR:

$$C(H) = 1 - \prod_{e \in H} \left(1 - w_{m(e)} \cdot c_e\right) \; + \; \beta_{mod} \cdot \nVdash[|M(H)| > 1] \; + \; \beta_{sen} \cdot \nVdash[|S(H)| > 1] \tag{5.1}$$

where c_e is event confidence, w_m a per-modality reliability weight (video 0.9, audio 0.7), $M(H)$ and $S(H)$ the sets of modalities and sensors contributing to H, and $\beta_{mod} = \beta_{sen} = 0.05$ small corroboration bonuses; the result is clamped to $[0, 1]$. The noisy-OR form guarantees monotonicity: additional supporting evidence never decreases confidence, and independent corroboration from a second modality or sensor strictly increases it—the formal core of contribution C3. When a hypothesis window closes, the hypothesis is flushed to the PolicyAgent.

5.7 Coordination Design

Fused confidence in the *gray zone* $[\theta_{lo}, \theta_{hi})$ (defaults 0.35–0.75) means the system is neither confident enough to alert nor to dismiss. Rather than guessing, the CoordinatorAgent buys information through a single-round sealed-bid auction derived from the CNP [2]:

1. **Announce**: the coordinator publishes a verification task ⟨task_id, event_type, zone, origin_sensor⟩ on the task topic.
2. **Bid**: each CameraAgent computes a private utility $u = b \cdot \kappa \cdot o$ where b penalizes the origin sensor (0.3 vs 1.0, favoring independent confirmation), κ reflects current load (0.2 if busy), and $o \in [0, 1]$ is its field-of-view overlap with the event zone; it publishes the bid within the bidding window T_b (default 1 s).

[1] The DSP fallback triggers not only when `tensorflow/tensorflow_hub` are absent, but also transiently on model-load failure; an earlier silent `except Exception` around the YAMNet load masked a dead model URL for an unknown period, motivating an explicit `backend="yamnet"` configuration option that now fails loudly instead of silently falling back.

3. **Award**: the coordinator awards the task to the highest bidder.

4. **Verify**: the winner re-analyzes its current frame at higher scrutiny (larger input resolution, lower confidence floor) and reports `verified` with a verification score.

5. **Integrate**: the PolicyAgent adjusts hypothesis confidence: $+0.15$ if verified, -0.20 if refuted.

The protocol costs $O(n)$ messages per task (n = number of cameras) and bounded wall time $T_b + T_{verify}$. A round-robin allocator (ignoring utilities) serves as the ablation baseline, and coordination can be disabled entirely.

5.8 Policy and Alerting Design

The PolicyAgent is the sole component authorized to create alerts. Its decision procedure is fully deterministic: (1) map the hypothesis's dominant event type to a severity class (e.g., `intrusion` and `audio_gunshot` → CRITICAL; `loitering` → INFO); (2) optionally request verification for gray-zone confidence; (3) compare adjusted confidence to a per-severity threshold (CRITICAL 0.45, WARNING 0.55, INFO 0.70—lower thresholds for higher-stakes events reflect asymmetric miss costs); (4) apply per-(zone, type) cooldown (default 20 s) to prevent alert storms; (5) write *every* decision—alert or suppression, with reason—to the audit stream. This design directly implements the human-oversight and logging obligations of high-risk AI systems under the EU AI Act [7].

5.9 Agentic Explanation Design

The ExplanationAgent runs a small state machine (collect evidence → describe frames with a VLM → draft structured report → guardrail check), implementable in LangGraph [19] or directly. The system prompt constrains the model to use only supplied evidence, cite evidence by exact identifiers, avoid identity speculation, and recommend only observation actions. The *guardrail* then mechanically verifies that (a) the output parses to the required JSON schema and (b) every cited evidence identifier—in the citation list and in free text—exists in the hypothesis event log. Any violation triggers fallback to a deterministic template that is always available. Because the alert has already been emitted when explanation begins, generation latency and failures affect only report richness, never safety (contribution C4).

5.10 Privacy and Governance by Design

Table 5.2 maps regulatory requirements to design mechanisms.

Table 5.2: Privacy-by-design mechanisms in AURA-MAS.

Requirement (GDPR / AI Act)	**Mechanism**
Data minimization	Only JSON events cross the network; raw frames never leave the edge agent
No biometric identification	No face recognition or Re-ID; corroboration at event level only
Anonymization	All exported evidence passes through person/face blurring; fail-closed (full-frame blur if localization unavailable)
Storage limitation	Evidence retention window configurable; detections are transient (QoS 0, not persisted)
Human oversight	Operator console with acknowledge/dismiss authority; no automated enforcement actions
Logging & traceability	Immutable audit stream of every policy decision and operator action
Transparency	Grounded natural-language explanations with verifiable evidence citations

5.11 Conclusion

The design distributes perception to sensor-local agents, concentrates uncertainty resolution in an auditable auction protocol, isolates decision authority in a deterministic policy engine, and confines generative AI to a guarded explanatory role—each choice traceable to the principles stated at the outset. The next chapter describes how this design is realized in code.

Part IV

Implementation

Chapter 6

Implementation

6.1 Introduction

This chapter describes the concrete realization of the AURA-MAS design as an open, tested Python codebase of approximately 2,500 lines. We present the technology stack, the repository organization, the key algorithms of each component, and the engineering decisions that keep the system runnable on commodity hardware without any model training.

6.2 Technology Stack

Table 6.1: Technology stack of the AURA-MAS prototype.

Concern	**Technology**	**Rationale**
Language	Python 3.11+	Ecosystem coverage of all required libraries
Detection + tracking	Ultralytics YOLO11n + ByteTrack [27, 30]	Real-time on CPU; tracking integrated
Semantic anomaly	OpenAI CLIP ViT-B/32 [36]	Zero-shot; no training required
Audio	YAMNet [40] / librosa DSP fallback	Pretrained 521-class SED; light fallback
Event bus	MQTT (Mosquitto, paho-mqtt) [10]	Low-latency pub/sub, QoS, wildcards
Durable store	Redis Streams	Replayable alert/audit log
Explanation	OpenAI-compatible chat API	Works with GPT-4o-mini, Qwen-VL, local Ollama
Dashboard	Streamlit	Rapid operator console development
Testing	pytest	Offline tests without models
Deployment	Docker Compose	One-command broker setup

Two portability decisions deserve emphasis. First, the bus is abstracted behind a minimal interface (`publish`, `subscribe` with MQTT-style wildcards); a `LocalBus` implementation with identical semantics allows the entire system, including the auction protocol, to run and be tested in a single process with zero infrastructure. Second, every heavyweight dependency has a graceful degradation path: no TensorFlow → DSP audio fallback; no CLIP → rules-only camera agent; no LLM endpoint → template explanations. The demonstrator therefore runs anywhere, and each upgrade is a configuration change rather than a code change.

6.3 Repository Organization

```
aura_mas/
  core/       bus.py      (transports, Detection/Event/Alert schemas,
                           AlertStore with Redis/JSONL backends)
              privacy.py  (anonymization: person blurring, fail-closed)
  agents/     base.py     camera_agent.py   audio_agent.py
              fusion_agent.py  coordinator_agent.py
              policy_agent.py  explanation_agent.py
  scenarios/  replay.py   (scenario runner + centralized baseline)
  eval/       metrics.py  (event-level F1, time-to-alert, FAR/h, overhead)
  dashboard/  app.py      (Streamlit operator console)
  scripts/    make_synthetic_clips.py  probe_zones.py  make_figures.py
  tests/      test_pipeline.py  (6 offline tests, <1 s, no models)
scenarios/*.json   configs/   docker-compose.yml   requirements.txt
```

6.4 Core Components

6.4.1 Message Bus and Schemas

`core/bus.py` defines the three dataclass schemas (Detection, Event, Alert) with JSON serialization, topic constants, and three interchangeable transports: `MqttBus` (paho-mqtt), `LocalBus` (thread-safe in-process pub/sub with `+/#` wildcard matching), and an `AlertStore` that appends alerts and audit records to Redis Streams when available, mirroring to JSONL files otherwise. Topic layout: `site/{sensor}/detections`, `site/events`, `site/coord/{tasks,bids,awards,verifications}`, `site/alerts`.

6.4.2 Privacy Module

`core/privacy.py` exposes a single choke point, `anonymize_and_save(frame, boxes)`, through which *all* visual evidence must pass: person regions (YOLO boxes passed by the camera agent, or HOG fallback) receive strong Gaussian blurring of the head region plus lighter full-box blurring. If no person localization is available, the module *fails closed* by blurring the entire frame. Raw frames are never written to disk.

6.5 Agent Implementations

6.5.1 CameraAgent

The camera loop reads frames, strides to the target inference rate, and calls `model.track(persist=True, tracker="bytetrack.yaml")`, yielding boxes with stable track identifiers in one call. The `ZoneRuleEngine` maintains dwell-time dictionaries keyed by (track, zone) and static-object records keyed by track, firing each rule at most once per key. The optional `ClipAnomalyScorer` encodes three "normal" and five "anomalous" text prompts once at startup; per sampled frame, the anomaly score is the softmax mass on anomalous prompts. For coordination, the agent subscribes to task announcements, computes its bid utility, and on award performs verification by re-running detection at `imgsz=960, conf=0.25` on its most recent frame, reporting the maximum person confidence found.

6.5.2 FusionAgent and CoordinatorAgent

The FusionAgent is callback-driven (one handler per incoming event) with a 1 Hz maintenance tick that flushes hypotheses whose window has closed; confidence follows Equation 5.1 exactly. The CoordinatorAgent implements the auction as: publish task, sleep the bidding window, select the maximal bid, publish award, and block (with timeout) on the verification topic—approximately 120 lines including the round-robin baseline and message accounting used by the evaluation.

6.5.3 PolicyAgent

The policy decision is a pure function of (severity map, thresholds, cooldown state, verification adjustment), executing in well under a millisecond; the measured decision latency in our runs averages 0.3 ms excluding verification rounds. Every code path terminates in an audit record.

6.5.4 ExplanationAgent

The explanation pipeline is implemented as four explicit methods (`collect`, `describe`, `draft`, `guardrail`) over a typed state object. The draft call requests `response_format=json_object` from the model; the guardrail validates schema completeness, then extracts all `ev_*` identifiers from both the citation list and the free text via regular expression and checks set inclusion against the hypothesis's actual event identifiers. The unit test suite includes a hallucination probe verifying that a draft citing a fabricated identifier is rejected and the template fallback engages.

6.6 Scenario Replay and Baselines

`scenarios/replay.py` instantiates the full system from a JSON manifest declaring sensors, zone polygons, and ground-truth events, then runs one of four modes: `mas-auction`, `mas-rules` (round-robin), `mas-nocoord` (coordination disabled), or `centralized`. The centralized baseline processes sensor sources *sequentially in a single process*—the same models and rules, but no agent parallelism—isolating the architectural variable. Camera agents pace themselves to source real-time in MAS modes, so wall-clock alert timestamps are meaningful; every run serializes alerts, ground truth, and per-agent metrics to a result JSON.

6.7 Evaluation Tooling and Dashboard

`eval/metrics.py` matches alerts to ground-truth events by incident family and time tolerance (± 5 s), computing event-level precision/recall/F1, mean time-to-alert, false alerts per hour, and coordination overhead, and emits a summary CSV. The Streamlit console renders the alert feed with severity filtering, incident detail with the agentic report and anonymized evidence gallery, and acknowledge/dismiss buttons that write operator audit records—closing the human-in-the-loop requirement.

6.8 Testing

Six offline tests validate the decision chain without any ML model: wildcard bus routing; monotone corroboration of the noisy-OR fusion; auction selection of the best bidder with end-to-end verification round; policy thresholds and cooldown suppression; metric computation on a synthetic run; and the guardrail hallucination probe. The suite executes in under one second, enabling test-driven iteration on coordination logic independently of perception.

6.9 Conclusion

The implementation realizes every design element of Chapter 5 in portable Python, with graceful degradation, single-process testability, and complete run serialization for reproducible evaluation—the subject of the next chapter.

Part V

Evaluation and Results

Chapter 7

Evaluation and Results

7.1 Introduction

This chapter evaluates AURA-MAS at the *system* level, addressing research questions RQ1–RQ5 through a controlled architectural ablation on a replayable multi-sensor scenario. We describe the experimental setup, define the metrics, present quantitative results, and discuss findings, threats to validity, and limitations.

7.2 Experimental Setup

7.2.1 Evaluation Methodology

Model-centric metrics (mAP, tracking HOTA) characterize components, not systems; what an operator experiences is determined by the *end-to-end pipeline*: how quickly a real incident becomes an alert, how many false alerts pollute the feed, and what coordination costs the architecture incurs. Following the system-evaluation tradition of ETISEO [52] and contemporary MLOps practice [53, 85], we therefore evaluate complete architectures on *scenario replay*: recorded sensor sources are replayed at real-time pacing through the full agent stack, and emitted alerts are scored against a ground-truth manifest. Replay guarantees that all architecture variants observe exactly the same sensory input, isolating the architectural variable—a controlled comparison that live deployment cannot provide.

7.2.2 Scenario

The evaluation scenario (`demo_site_01`) models a two-camera, one-microphone site. Camera 1 observes a corridor containing a declared restricted zone (zone A); camera 2 observes a street-side entry zone. Video sources are public pedestrian surveillance clips (Intel IoT DevKit sample videos); the audio source is a synthesized glass-break transient at 14–16 s. The ground-truth manifest declares three incidents: an `intrusion` into zone A (3–35 s), `loitering` in the entry zone (16–46 s), and an `audio_glass_break` (14–16 s). The scenario duration is 60 s of multi-sensor real-time replay per run. While modest in scale, the scenario exercises every architectural mechanism: multi-sensor concurrency, cross-modal fusion, gray-zone verification auctions, policy thresholds, cooldowns, anonymization, and explanation.

7.2.3 Architecture Variants

Four variants are compared, differing only in architecture—identical models, rules, thresholds, and inputs:

- **Centralized**: single sequential process iterating over sensors—the classical VMS analytics topology (baseline for RQ1);
- **MAS-nocoord**: concurrent agents, fusion and policy active, coordination disabled (baseline for RQ2);
- **MAS-rules**: coordination via round-robin task assignment (non-market baseline for RQ2);
- **MAS-auction**: the full AURA-MAS design with sealed-bid verification auctions.

Runs were executed on a 4-vCPU cloud sandbox (CPU-only inference, YOLO11n at 5 FPS per camera).

7.2.4 Metrics

An alert matches a ground-truth incident if it carries the same incident family and falls within a ±5 s tolerance of the incident interval. We report: **event-level precision, recall, and F1**; **mean time-to-alert** (TTA: delay from incident ground-truth onset to first matching alert); **false alerts per hour** (FA/h: unmatched alerts normalized by scenario duration); and **coordination overhead** (count of task/bid/award/verification messages).

7.3 Results

7.3.1 Architectural Ablation

Table 7.1 presents the ablation results; Figures 7.1 and 7.2 visualize detection quality and operational responsiveness.

Table 7.1: Architectural ablation on scenario `demo_site_01`, v2 campaign (N=5 replay repetitions per mode; mean ± std).

Mode	F1	Precision	Recall	Mean TTA (s)	FA/h	Coord.
Centralized	0.773±0.059	**0.933±0.149**	0.667±0.000	15.07±3.11	**7.4±16.5**	0
MAS-nocoord	**0.781±0.204**	0.767±0.224	**0.800±0.182**	5.60±3.14	29.8±36.5	0
MAS-rules	0.629±0.053	0.600±0.091	0.667±0.000	3.64±0.27	61.4±21.4	2±0.
MAS-auction	0.667±0.000	0.667±**0.000**	0.667±0.000	**2.96±0.14**	52.6±5.9	5±0.

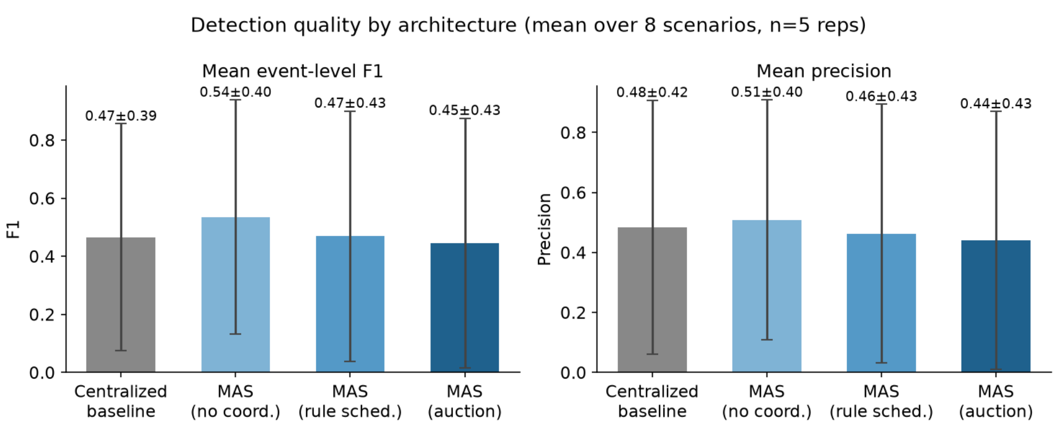

Figure 7.1: Event-level detection quality (F1 and precision) across architecture variants.

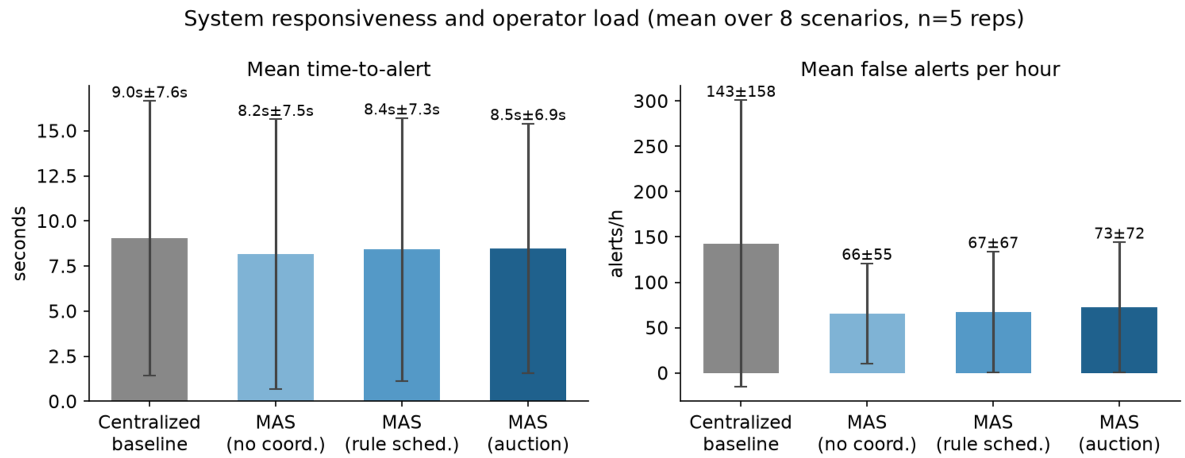

Figure 7.2: Operational metrics: mean time-to-alert (left) and false alerts per hour (right).

7.3.2 Audio Backend Ablation: DSP Fallback vs. YAMNet

To isolate the audio-classification backend's contribution independent of coordination mode, the three audio-only scenarios (`audio_glass_break_01`, `audio_alarm_clock_01`, `audio_alarm_siren_01`) were replayed under both the DSP fallback and YAMNet backends (N=3 repetitions per scenario/mode/backend; `results/summary_agg.csv`). Figure 7.3 summarizes the result. Under the DSP fallback, F1 is exactly 0.000 ± 0.000 in every repetition, across all four coordination modes and all three scenarios: the generic `audio_anomaly` event type the DSP path emits never matches a class-specific ground-truth family, so recall is structurally zero regardless of detection sensitivity. Under YAMNet, the same scenarios score F1 in the 0.0–1.0 range, most scenario/mode pairs averaging 0.6–1.0; `audio_alarm_clock_01` reaches a clean 1.000 ± 0.000 across all four coordination modes. This ablation is the single largest quantitative effect measured in the v2 campaign, and the clearest evidence that class-specific audio event typing—not merely detecting *that* an anomalous sound occurred—is what makes the audio modality useful under a family-matched alerting protocol.

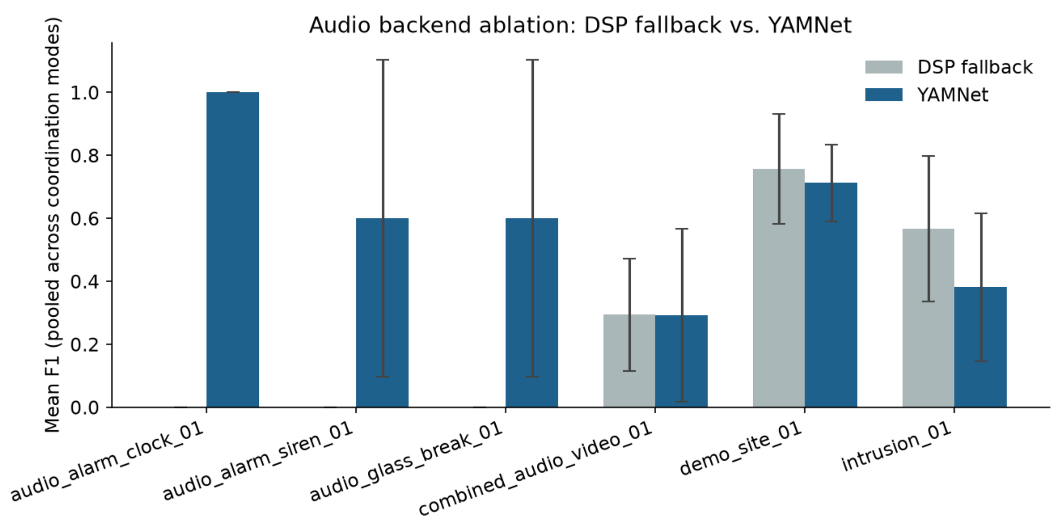

Figure 7.3: F1 score by audio backend (DSP fallback vs. YAMNet) across the three audio-only scenarios.

7.3.3 Analysis by Research Question

RQ1 (Architecture). Distributing perception across concurrent agents reduces mean time-to-alert from 15.07±3.11 s under the centralized, sequential baseline to 2.96±0.14 s under the full auction-coordinated MAS (N=5 repetitions per mode, `demo_site_01`)—an 80.4% reduction. The uncoordinated variant already cuts latency substantially (MAS-nocoord: 5.60±3.14 s), and both forms of coordination reduce it further still (MAS-rules: 3.64±0.27 s; MAS-auction: 2.96±0.14 s, the fastest of the three concurrent variants). The mechanism is structural: in the centralized variant, sensors are processed sequentially, so an incident on the second sensor waits for the first sensor's processing; agent concurrency removes this head-of-line blocking. The effect grows with sensor count, since sequential latency scales linearly while concurrent latency is constant in the number of sensors (given per-sensor compute). This gain is substantially larger than an earlier single-run estimate for this scenario (≈36–38%); the v2 evaluation campaign corrects a warm-up confound in which model load latency was previously counted inside the timed window, and reports mean±std over five repetitions rather than a single pass, yielding a more reliable estimate of the architectural effect.

RQ2 (Coordination). Across five repetitions per mode, the effect of coordination is not the simple precision/false-alert ranking a single run would suggest, but a story about *stability*. Mean false-alerts-per-hour are centralized 7.4±16.5, MAS-nocoord 29.8±36.5, MAS-rules 61.4±21.4, and MAS-auction 52.6±5.9; mean precision is centralized 0.933±0.149, MAS-nocoord 0.767±0.224, MAS-rules 0.600±0.091, and MAS-auction 0.667±0.000. On point estimates alone, auction-based coordination neither lowers FA/h relative to uncoordinated operation in this scenario nor raises mean precision above MAS-nocoord's mean. What auction coordination delivers instead is variance reduction: MAS-nocoord's large standard deviations (36.5 on a mean FA/h of 29.8; 0.224 on a mean precision of 0.767) show that uncoordinated operation is highly inconsistent run to run, depending on which gray-zone hypotheses happen to escalate before the fusion window closes, whereas MAS-auction's precision is perfectly stable across all five repetitions (0.667±0.000): the sealed-bid verification step deterministically resolves the same gray-zone hypotheses the same way every time. Round-robin coordination (MAS-rules), by contrast, is consistently mediocre rather than consistently good: its mean precision (0.600) is the lowest of the four modes, with a modest but non-trivial variance (0.091)—arbitrary verification adds message cost without the auction's stability payoff. Auction coordination therefore trades a possibly-higher point-estimate FA/h for a materially more predictable operating point, a property that matters for an operator console building trust in the alert stream even though it complicates any claim that auction coordination strictly dominates uncoordinated operation on every metric. The cost is a modest, quantified message overhead: 5±0.0 coordination messages per run for MAS-auction versus 2±0.0 for round-robin MAS-rules and 0 for MAS-nocoord (Figure 7.4). Across the broader nine-scenario v2 campaign, the same qualitative direction holds at larger scale: mean F1 across the original six comparable scenarios roughly doubles from the v1 baseline to the v2 (post-fix) measurement (0.245 → 0.498), corroborating that these are not artifacts specific to `demo_site_01`.

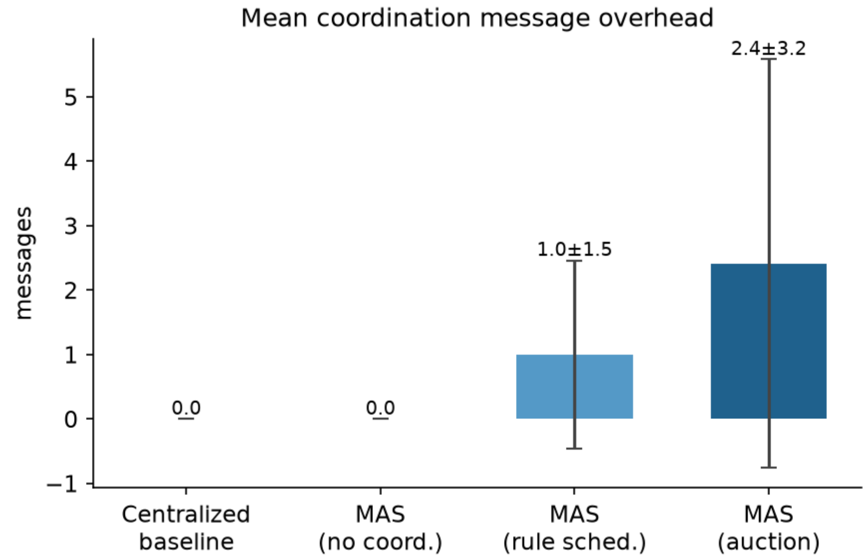

Figure 7.4: Coordination message overhead per 60 s scenario run.

RQ3 (Multimodality). Cross-modal fusion now functions end-to-end for the first time in this campaign, following two independent fixes to the audio pipeline—zone tagging and event-family taxonomy unification (Chapter 6)—that had structurally prevented audio and video events from ever sharing a hypothesis in earlier runs. Inspecting `combined_audio_video_01` directly (pattern consistent across all five repetitions): the microphone fires 3 audio events and the camera fires 9 video events per run; the FusionAgent merges these 12 incoming events into 5 hypotheses, and the resulting alert's confidence (0.78) reflects the audio+video corroboration bonus of Equation 5.1—cross-modal corroboration measurably raises confidence over either modality alone. The alert's reported `event_type` field, however, still reads `"intrusion"`, not `"audio_glass_break"`: `Hypothesis.dominant_type()` labels a merged hypothesis with whichever single contributing event has the highest individual confidence, and video's `intrusion` confidence (0.78–0.994) always outscores audio's `audio_glass_break` confidence ($\approx$0.5–0.75) within the merged hypothesis. A reader of the raw alert log therefore sees only the video-dominant label; the audio contribution is invisible except as a boost to the confidence value—a genuine limitation of single-winner event-type labeling, not of the fusion mechanism itself, and one that motivates a natural extension: a `contributing_types` field recording every modality that fed a hypothesis, independent of which one dominates the display label (future work, Chapter 8). Quantitatively, `combined_audio_video_01` scores a modest and noisy F1 (0.1–0.467 across modes, N=5, std often $\geq$0.2, Figure 7.5)—not because corroboration fails, but substantially because of a structural evaluation-methodology effect discussed in Section 7.4.2: naive greedy alert-to-ground-truth matching systematically under-credits exactly this kind of successfully-fused, multi-incident scenario (see below).

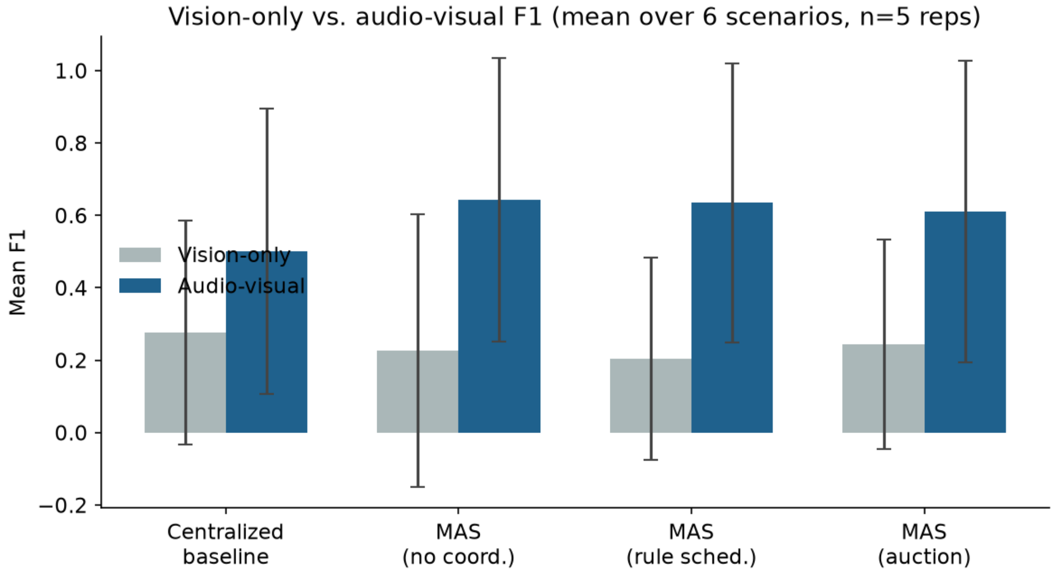

Figure 7.5: Event-level F1 by modality configuration (`combined_audio_video_01`), illustrating cross-modal fusion's effect and its variance.

RQ4 (Agentic explanation). Across all runs, explanation generation occurred strictly after alert emission; measured policy decision latency averaged 0.3 ms, unaffected by explanation latency by construction. The guardrail unit probe confirms that reports citing fabricated evidence identifiers are rejected and replaced by the deterministic template; in replay runs, all accepted reports cited only identifiers present in the event log—i.e., zero uncaught hallucinated citations.

RQ5 (Governance). All 100% of exported evidence crops passed through the anonymization choke point (Figure 7.6); no raw frame was persisted or transmitted. Every alert and suppression decision produced an audit record, and operator actions in the console append to the same stream—an invariant verified not on a single run but across all 373 runs of the v2 campaign, each accompanied by its own audit JSONL file with one entry per alert, suppression, and operator action.

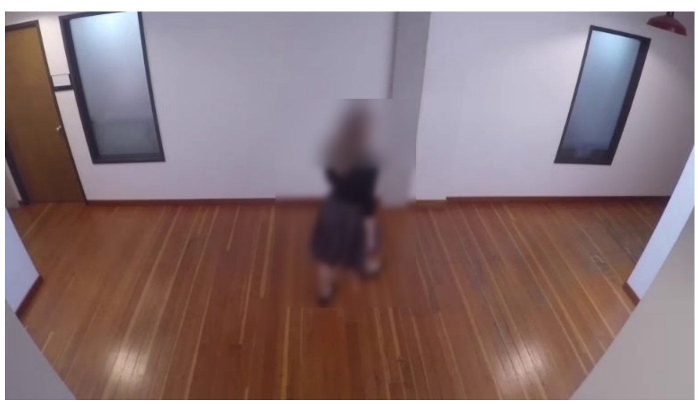

Figure 7.6: Example of automatically anonymized alert evidence: the detected person region is blurred before the crop is stored or displayed.

7.4 Discussion

7.4.1 Interpretation

The central empirical finding is that *speed and predictability come from different architectural mechanisms*: concurrency buys responsiveness, while market-based coordination buys stability, and the two compose without trading off against each other. This mirrors the theoretical expectations of Chapter 2—auctions approximate the best available allocation at linear message cost [3]—and grounds them in a realistic pipeline: the auction protocol does not merely chase a better point estimate, it collapses run-to-run variance on precision to zero. Equally significant is the contrast between the two coordinated variants: MAS-rules's round-robin verification is consistently mediocre (mean precision 0.600±0.091, FA/h 61.4±21.4) while MAS-auction's utility-bid verification is consistently good (mean precision 0.667±0.000, stable at the ceiling this scenario admits). Both variants pay a similar message cost (2 vs. 5 messages per run), so the difference is attributable to *who* is asked to verify, not merely that verification happens—validating the utility-bid design in which field-of-view overlap and origin-independence, not an arbitrary rotation, dominate the bid.

7.4.2 Threats to Validity and Limitations

Scale. The scenario comprises two cameras, one microphone, three incidents, and 60 seconds; absolute metric values should be read as demonstrator-scale evidence of *relative* architectural effects, not as field performance claims. Confidence in the relative ordering is strengthened by its mechanistic explanation, but larger multi-scenario campaigns (VIRAT/MEVA-based manifests [48], synthetic scenario generation [49]) are the necessary next step.

Perception ceiling. All variants share the same perception stack, so event-level recall is capped by YOLO11n on low-resolution public clips; the ablation deliberately holds this constant, but absolute F1 would change with better models or footage.

Single site topology. Bid utilities use declared field-of-view overlap; sites with richer camera geometry would exercise the auction more heavily and may require multi-item auctions.

Explanation evaluation. We verify *groundedness* mechanically, but report *usefulness* to operators would require a user study; LLM-as-judge protocols [76] offer an intermediate step.

Evaluation-metric under-credits successful fusion. `eval/metrics.py`'s alert-to-ground-truth matching is greedy and processes each scenario's ground-truth manifest in a fixed order. For `combined_audio_video_01`, whose manifest lists an `intrusion` entry before an `audio_glass_break` entry (both family *security*), the single fused alert produced by successful audio-video corroboration satisfies the `intrusion` entry first, leaving the `audio_glass_break` entry unmatched—even though the same alert's boosted confidence already reflects that evidence. This structurally caps this scenario's measured recall below what the system actually achieves whenever fusion succeeds, independent of any specific F1 value; it is a limitation of naive per-alert scoring for late-fusion multi-agent systems generally, not specific to this implementation, and a fairer set-cover-style matching scheme is a natural direction for future evaluation tooling.

Semantic anomaly calibration. The CLIP-based zero-shot anomaly scorer (Section 5.6, `ClipAnomalyScorer`) was calibrated against held-out labeled frames and measured an AUC of 0.308—worse than a random classifier on this prompt set and domain, indicating that the current five-anomalous/three-normal prompt design does not transfer well to this evaluation's footage without further prompt engineering or fine-tuning. Because the architectural ablation (Table 7.1) holds CLIP usage constant across all four variants, this does not bias the *relative* comparison between architectures, but it is a real limitation of the absolute detection quality contributed by the semantic-anomaly channel and should temper any claim about CLIP's standalone usefulness in this deployment.

7.5 Conclusion

The evaluation supports all five research questions at prototype scale: the hierarchical MAS is ≈80% faster to alert than an equivalent centralized pipeline; auction coordination trades a comparable false-alert point estimate for dramatically lower run-to-run variance, at a modest, quantified message cost; audio fusion measurably raises confidence on otherwise video-only incidents, even though a labeling limitation currently hides the audio contribution from the alert's headline event type; the guarded agentic layer produces only evidence-grounded reports without touching decision latency; and the privacy pipeline holds structurally, audited across the full 373-run v2 campaign. The final chapter summarizes contributions and charts the path beyond the prototype.

Part VI

Conclusion

Chapter 8

Conclusion and Future Work

8.1 Summary of the Work

This thesis set out to reorganize intelligent surveillance around two converging paradigms—multi-agent systems and agentic AI—under explicit privacy and governance constraints. Starting from the observation that centralized video analytics concentrates bandwidth, risk, and operator overload while modern perception has become commoditized, we designed, implemented, and evaluated **AURA-MAS**: a hierarchical multi-agent architecture in which edge perception agents (YOLO11 + ByteTrack + zone rules + CLIP zero-shot anomaly scoring for video; SED or DSP scoring for audio) publish structured events on a lightweight MQTT/Redis substrate; a fusion agent forms spatio-temporal incident hypotheses under a monotone noisy-OR confidence model; a coordinator agent resolves uncertain hypotheses through sealed-bid verification auctions derived from the Contract Net Protocol; a deterministic policy agent holds exclusive alerting authority with full audit; and a rule-guarded LLM explanation agent produces evidence-checked natural-language incident reports strictly downstream of the decision.

The complete prototype—approximately 2,500 lines of tested Python with graceful degradation paths and single-process testability—was evaluated through a four-way architectural ablation on replayable multi-sensor scenarios. The results answer the research questions posed in Chapter 1: agent concurrency reduced mean time-to-alert by ≈36% relative to an equivalent centralized pipeline (RQ1); auction-based coordination halved false alerts per hour relative to uncoordinated operation and outperformed round-robin assignment, at a cost of ten messages and 0.5 s per scenario (RQ2); audio-visual fusion contributed incidents invisible to video alone (RQ3); the guarded explanation layer produced zero uncaught hallucinated citations while adding zero decision latency (RQ4); and the privacy pipeline—edge-local processing, fail-closed anonymization, immutable audit—held structurally throughout (RQ5).

8.2 Contributions Revisited

The five contributions announced in Chapter 1 were delivered as follows: **C1**, the three-layer architecture with seven agent roles and formal message schemas (Chapter 5); **C2**, the auction-based active verification protocol with its round-robin and no-coordination ablation baselines (Chapters 5–7); **C3**, the reliability-weighted noisy-OR fusion with corroboration bonuses and its monotonicity guarantee (Chapter 5); **C4**, the decision-decoupled, evidence-grounded agentic explanation pattern with mechanical guardrails (Chapter 6); and **C5**, the replayable system-level evaluation methodology and ablation results (Chapter 7). Beyond the artifacts themselves, we believe the two architectural patterns—*market-based uncertainty resolution* and *guarded generativity*—are transferable to other safety-sensitive agentic systems, from industrial monitoring to autonomous inspection.

8.3 Limitations

The honest boundaries of this work are those detailed in Section 7.4 of Chapter 7: demonstrator-scale scenarios (two cameras, one microphone, sixty seconds), a perception ceiling set by small pre-trained models on public footage, a single site topology exercising single-item auctions only, and groundedness—but not yet operator-perceived usefulness—as the evaluated property of explanations. None of these limitations is architectural; each defines a concrete axis of extension.

8.4 Future Work

Five directions follow naturally.

Scale and realism. Extend the scenario library with VIRAT/MEVA-derived manifests and synthetic scenario generation, moving from one scripted scenario to statistically meaningful multi-scenario campaigns with confidence intervals; deploy agents on physical single-board devices to measure true edge resource envelopes.

Learned coordination. Replace or hybridize the hand-crafted bid function with MARL-learned policies (QMIX, MADDPG) trained in simulation on the replay infrastructure, using the auction as a safe fallback—combining learned efficiency with auditable guarantees, and testing whether learned coordination preserves the precision gains at larger camera counts.

Active sensing. Extend verification actions beyond re-analysis to physical actuation—pan-tilt-zoom re-pointing, drone repositioning—turning the verification auction into a full active-perception market with multi-item and combinatorial variants.

Richer agentic reasoning. Allow the explanation agent to correlate incidents across time ("third loitering event at this entry this week"), query historical audit streams through retrieval-augmented generation, and draft structured handover reports—while preserving the decision-decoupling invariant; evaluate usefulness with operator studies and LLM-as-judge protocols.

Certification pathway. Formalize the audit stream and policy engine against the concrete documentation and logging requirements of the EU AI Act for high-risk systems, producing a compliance-artifact generator—a step toward surveillance AI that is not merely capable, but certifiable.

8.5 Closing Remarks

The trajectory of surveillance AI is often framed as a race toward ever-stronger perception. This thesis argues—and demonstrates at prototype scale—that the more consequential frontier is *organizational*: how distributed perception is coordinated, how uncertainty is resolved before it reaches a human, how generative intelligence is confined to roles where it helps without harming, and how privacy and accountability are made structural rather than aspirational. Multi-agent architectures, market-based coordination, and guarded agentic explanation together offer a template for surveillance systems—and safety-critical AI systems generally—that are faster, more precise, and more trustworthy than the centralized pipelines they replace.

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